

Technical Documentation: Smart Building AI

1. Introduction

Smart Building AI is an autonomous, closed-loop Building Management System (BMS) designed to optimize building energy consumption while maintaining occupant thermal comfort. Unlike conventional schedule-based HVAC control systems, this platform uses a Large Language Model (LLM) as an intelligent decision engine that continuously analyzes real-time building conditions and determines optimal control strategies.

The system integrates directly with an EnergyPlus building simulation, allowing it to monitor environmental conditions, reason about energy efficiency opportunities, and automatically adjust HVAC setpoints. Every control decision is validated through predefined safety constraints before being applied, ensuring reliable and stable operation.

2. System Architecture

The platform follows a modular, decoupled architecture that separates simulation, artificial intelligence, data management, and visualization into independent layers.

Simulation Layer

The simulation layer is powered by **EnergyPlus**, which acts as the digital representation of the building. Rather than running as a static simulation, EnergyPlus executes incrementally, allowing the backend to interact with the simulation after every timestep.

Backend Layer

The backend is implemented using **FastAPI** and coordinates communication between the EnergyPlus simulator, the AI decision engine, the PostgreSQL database, and the frontend dashboard. It exposes REST APIs for simulation control, data retrieval, and AI interaction.

AI Decision Layer

The intelligence of the system is provided by a locally hosted **Mistral 7B** model running through **Ollama**. At each simulation interval, the AI receives current building telemetry, evaluates the environmental conditions, and generates Energy Conservation Measures (ECMs) aimed at minimizing energy consumption without compromising occupant comfort.

Data Layer

A PostgreSQL database stores all simulation metadata, sensor readings, AI reasoning, and applied HVAC control actions. This enables historical analysis, auditing, and performance evaluation.

Presentation Layer

A React-based dashboard provides operators with real-time visualization of energy usage, environmental conditions, AI recommendations, historical trends, and simulation status.

3. Technology Stack

Frontend

- React
- Vite
- Tailwind CSS
- Recharts
- React Query

Backend

- Python 3.14
- FastAPI
- SQLAlchemy (Async)
- Alembic

Database

- PostgreSQL
- NeonDB (Production)
- Local PostgreSQL (Development)

Artificial Intelligence

- Mistral 7B
- Ollama
- MCP Server

Simulation

- EnergyPlus v26.2.0
- PyEnergyPlus API

4. Database Design

The database schema is designed to efficiently manage high-frequency simulation data while maintaining complete traceability of AI decisions.

simulations

Stores metadata for every simulation execution, including simulation status, start time, end time, execution mode, and overall performance metrics.

sensor_logs

Records time-series telemetry generated during simulation, including indoor temperature, outdoor temperature, humidity, HVAC power consumption, total building energy usage, and simulation timestep.

ai_decisions

Maintains a complete audit trail of every LLM inference, including the prompt context, generated reasoning, confidence level, and recommended control strategy.

control_actions

Stores every HVAC adjustment applied to the building simulation, such as cooling setpoint modifications, heating setpoint changes, fan speed adjustments, and timestamps.

This normalized schema enables efficient historical analysis while supporting thousands of sensor updates during long simulation runs.

5. AI Decision Engine

The Smart Building AI employs a locally hosted **Mistral 7B** Large Language Model running through Ollama, eliminating the need for external API services while ensuring complete data privacy.

The AI control process consists of the following stages:

1. Current environmental conditions are collected from the EnergyPlus simulation.
2. Sensor data is formatted into a deterministic prompt containing building state, operational constraints, and optimization objectives.
3. The Mistral model analyzes the current conditions and generates recommended Energy Conservation Measures (ECMs).
4. A safety validation module verifies that all recommendations satisfy predefined operational limits.
5. Approved control actions are forwarded to the EnergyPlus simulation for execution.

Typical AI recommendations include:

- Increasing cooling setpoint during low occupancy.
- Reducing HVAC operation during mild weather.
- Maintaining indoor temperatures within occupant comfort limits.
- Minimizing unnecessary energy consumption.

6. EnergyPlus Integration

The project integrates directly with **EnergyPlus** using the **PyEnergyPlus API**, enabling real-time bidirectional communication with the building simulation.

At each simulation timestep, the backend can:

- Pause simulation execution.
- Read internal building variables.
- Extract environmental sensor data.

- Modify HVAC actuator values.
- Resume simulation execution.

This event-driven interaction transforms EnergyPlus into an interactive digital twin instead of a traditional offline simulator.

7. Closed-Loop Communication Flow

The system operates as a sequential autonomous control loop.

Step 1: EnergyPlus advances one simulation timestep.

Step 2: Sensor Collector retrieves updated environmental variables through the PyEnergyPlus API.

Step 3: FastAPI stores the collected telemetry inside PostgreSQL.

Step 4: The Mistral AI agent receives the latest building state and determines an optimized HVAC control strategy.

Step 5: The Decision Validator checks that all proposed actions remain within predefined safety and comfort constraints.

Step 6: Validated control actions are injected into the EnergyPlus simulation.

Step 7: The simulation resumes and the cycle repeats until completion.

This closed-loop architecture enables continuous optimization throughout the entire simulation.

8. REST API Endpoints

Simulation Management

POST /simulation/start

- Starts a new EnergyPlus simulation.

POST /simulation/pause

- Temporarily pauses the running simulation.

POST /simulation/stop

- Gracefully terminates the simulation.

GET /simulation/status

- Returns the current simulation state and execution progress.
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Sensor APIs

GET /sensor/live

- Retrieves the latest sensor telemetry from the simulation.

GET /sensor/history

- Returns historical environmental and energy data for visualization and trend analysis.
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AI Decision APIs

POST /agent/decide

- Manually triggers an AI decision cycle.

GET /agent/history

- Retrieves previous LLM reasoning logs and applied HVAC control actions.
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Dashboard APIs

GET /dashboard/summary

- Returns high-level KPIs including current energy consumption, indoor conditions, and AI performance metrics.

GET /dashboard/charts

- Returns formatted time-series datasets for graphical visualization using Recharts.
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9. Key Features

- Autonomous closed-loop HVAC control
 - Local LLM inference using Mistral 7B through Ollama
 - Real-time EnergyPlus simulation integration
 - Safety-constrained decision validation
 - High-frequency sensor telemetry storage
 - Historical AI reasoning and action auditing
 - Interactive React dashboard with live monitoring
 - Modular FastAPI backend architecture
 - PostgreSQL-based persistent data storage
 - Fully extensible REST API for future integrations
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10. Conclusion

Smart Building AI demonstrates how modern Large Language Models can be integrated into Building Management Systems to create intelligent, autonomous energy optimization platforms. By combining EnergyPlus simulation, FastAPI, PostgreSQL, and a locally hosted Mistral 7B model, the

system forms a complete closed-loop architecture capable of monitoring building conditions, generating optimized HVAC control strategies, validating decisions through safety constraints, and continuously improving operational efficiency.

The modular architecture allows future enhancements such as occupancy prediction, renewable energy integration, reinforcement learning, demand response optimization, and deployment on real-world smart buildings.